

Oracle RAC 19c on VMware ESXi 6.7 ??? Complete Build & Administration Guide

2-Node Cluster: racnode1 / racnode2

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1. Architecture Overview

1.1 Environment Summary

Item	Value
Hypervisor	VMware ESXi 6.7.0
Nodes	racnode1, racnode2
OS	Oracle Linux 7.9 x86_64
RAM per node	8 GB (upgraded from initial 6 GB)
OCPU per node	Maximum available
Grid Infrastructure	Oracle 19.3 + RU 39036936
Database	Oracle 19.3 + RU 39036936, RAC, Admin-Managed, CDB with 1 PDB
Cluster Name	racnode-cluster
Database Name / SID Prefix	racdb (racdb1 / racdb2)
ASM Disk Groups	GRID (16 GB, OCR/Voting), DATA (50 GB, DB files)
ASM Redundancy	External (single-disk, no OS mirroring)
Storage source	Single ESXi datastore, shared via multi-writer flag

1.2 Network Layout

Two VMware Standard Port Groups on vSwitch0:

- **VM Network** → Public interface (ens192)
- **RAC_Network** → Private interconnect (ens224)

Both port groups share the same vSwitch, so interconnect traffic between nodes never leaves the ESXi host — no physical uplink needed for RAC_Network.

1.3 IP Address Plan (Updated — 9 IPs total)

Purpose	Hostname	IP Address	Subnet
racnode1 (public)	racnode1.local	192.168.120.240	192.168.120.0/24

Purpose	Hostname	IP Address	Subnet
racnode2 (public)	racnode2.local	192.168.120.241	192.168.120.0/24
racnode1-vip	racnode1-vip.local	192.168.120.242	192.168.120.0/24
racnode2-vip	racnode2-vip.local	192.168.120.243	192.168.120.0/24
racscan (SCAN #1)	racscan.local	192.168.120.244	192.168.120.0/24
racscan (SCAN #2)	racscan.local	192.168.120.247	192.168.120.0/24
racscan (SCAN #3)	racscan.local	192.168.120.249	192.168.120.0/24
racnode1-priv	racnode1-priv.local	192.168.32.235	192.168.32.0/24
racnode2-priv	racnode2-priv.local	192.168.32.236	192.168.32.0/24

Netmask: 255.255.255.0 on both interfaces. No gateway/DNS on the private (32.x) interface — must stay non-routable.

Note on 3 SCAN IPs with /etc/hosts only: With no real DNS server, /etc/hosts can still return multiple IPs for a single name by listing three separate lines for racscan. Linux's resolver will hand back all three addresses to Grid Infrastructure's SCAN listener setup. This satisfies Oracle's 3-SCAN-IP best practice (better client load balancing and faster failover) even without DNS round robin.

/etc/hosts (identical on both nodes)

```
192.168.120.240 racnode1.local racnode1
192.168.120.241 racnode2.local racnode2
192.168.120.242 racnode1-vip.local racnode1-vip
192.168.120.243 racnode2-vip.local racnode2-vip
192.168.120.244 racscan.local racscan
192.168.120.247 racscan.local racscan
192.168.120.249 racscan.local racscan
192.168.32.235 racnode1-priv.local racnode1-priv
192.168.32.236 racnode2-priv.local racnode2-priv
```

2. ESXi VM Configuration

2.1 Port Groups (already existing, confirmed)

Host Client → Networking → Port groups: VM Network, RAC_Network, both on vSwitch0, VLAN 0.

2.2 VM Creation — Network Adapters

Per VM (racnode1, racnode2), two vNICs:

Adapter	Port Group	Type
Network Adapter 1	VM Network	VMXNET3

Adapter	Port Group	Type
Network Adapter 2	RAC_Network	VMXNET3

Both set to **Connect at power on**. Keep adapter order identical on both nodes.

2.3 VM Sizing (Updated)

- **RAM: 8 GB per node** (updated from 6 GB — meets Oracle's official 19c GI minimum, removes the need for memory prerequisite overrides)
- vCPU: maximum available on host
- Swap: recommend ≥ 8 GB, consistent across both nodes

2.4 Disks

Disk	Size	Controller	Sharing	Purpose
Hard disk 1 (OS)	100 GB	SCSI 0 (Paravirtual, Bus Sharing = None)	Not shared	OS boot, per-node
Hard disk 2 (GRID)	16 GB	SCSI 1 (Paravirtual, Bus Sharing = Virtual)	Shared, Thick Eager Zeroed	OCR + Voting
Hard disk 3 (DATA)	50 GB	SCSI 1 (Paravirtual, Bus Sharing = Virtual)	Shared, Thick Eager Zeroed	Database files

Steps (racnode1 first)

1. Edit Settings → Add other device → SCSI Controller → Type: VMware Paravirtual → SCSI Bus Sharing: **Virtual**
2. Add hard disk → New standard hard disk → 16 GB → expand → Disk Provisioning: **Thick provision eager zeroed** → Virtual Device Node: SCSI controller 1, slot (1:0)
3. Repeat for 50 GB disk → slot (1:1)
4. Save

Enable multi-writer flag (SSH into ESXi host, VM powered off)

```
cd /vmfs/volumes/<datastore>/racnode1/
vi racnode1.vmx
```

Add:

```
scsi1.sharedBus = "virtual"
scsi1:0.sharing = "multi-writer"
scsi1:1.sharing = "multi-writer"
```

Attach same disks to racnode2

Edit Settings on racnode2 (powered off) → Add SCSI Controller 1 (Paravirtual, Virtual) → Add hard disk → **Existing hard disk** → browse to racnode1's folder → select the same GRID/DATA .vmdk files → same slots (1:0), (1:1) → Save → edit racnode2.vmx with the same 3 multi-writer lines → power on.

3. Guest OS Network Configuration

Identify interfaces:

```
nmcli device status
ip a
```

ens192 = public, ens224 = private (confirm via MAC address matching the vSphere adapter order).

racnode1

```
nmcli con mod ens192 ipv4.method manual ipv4.addresses 192.168.120.240/24 \
  ipv4.gateway 192.168.120.1 ipv6.method disabled
nmcli con up ens192
```

```
nmcli con add type ethernet ifname ens224 con-name ens224 \
  ipv4.method manual ipv4.addresses 192.168.32.235/24 \
  ipv6.method disabled connection.autoconnect yes
nmcli con up ens224
```

racnode2

```
nmcli con mod ens192 ipv4.method manual ipv4.addresses 192.168.120.241/24 \
  ipv4.gateway 192.168.120.1 ipv6.method disabled
nmcli con up ens192
```

```
nmcli con add type ethernet ifname ens224 con-name ens224 \
  ipv4.method manual ipv4.addresses 192.168.32.236/24 \
  ipv6.method disabled connection.autoconnect yes
nmcli con up ens224
```

No gateway/DNS on ens224 (private interconnect) — must remain non-routable.

Add the /etc/hosts block from Section 1.3 to **both** nodes.

4. Shared Storage (ASM Disks)

4.1 Persistent Device Naming (udev)

Get WWIDs (from racnode1, both disks visible identically on racnode2 due to multi-writer sharing):

```
for d in sdb sdc; do
  echo "$d: $(/usr/lib/udev/scsi_id -g -u -d /dev/$d)"
done
```

Create /etc/udev/rules.d/99-oracle-asmdevices.rules identically on **both nodes**:

```
KERNEL=="sd*", SUBSYSTEM=="block", ENV{ID_SERIAL}=="<GRID_WWID>",
SYMLINK+="asm-grid01", OWNER="grid", GROUP="asmadmin", MODE="0660"
```

```
KERNEL=="sd*", SUBSYSTEM=="block", ENV{ID_SERIAL}=="<DATA_WWID>",  
SYMLINK+="asm-data01", OWNER="grid", GROUP="asmadmin", MODE="0660"
```

Reload:

```
udevadm control --reload-rules  
udevadm trigger --type=devices --action=change  
ls -l /dev/asm-grid01 /dev/asm-data01
```

Ownership resolves to grid:asmadmin only after the grid user and asmadmin group exist (Section 5).

5. OS Preparation (both nodes, identical)

5.1 Groups and Users

```
groupadd -g 54321 oinstall  
groupadd -g 54322 dba  
groupadd -g 54323 oper  
groupadd -g 54324 asmadmin  
groupadd -g 54325 asmdba  
groupadd -g 54326 asmoper  
groupadd -g 54327 racdba
```

```
useradd -u 54321 -g oinstall -G asmadmin,asmdba,asmoper,dba,racdba grid  
useradd -u 54322 -g oinstall -G dba,oper,asmdba,racdba oracle
```

```
passwd grid  
passwd oracle
```

5.2 Directories

```
mkdir -p /u01/app/grid  
mkdir -p /u01/app/19.0.0/grid  
mkdir -p /u01/app/oracle/product/19.0.0/dbhome_1  
mkdir -p /u01/app/oraInventory  
chown -R grid:oinstall /u01  
chown -R oracle:oinstall /u01/app/oracle  
chown -R grid:oinstall /u01/app/oraInventory  
chmod -R 775 /u01
```

5.3 Central Inventory Pointer

```
cat > /etc/oraInst.loc << 'EOF'  
inventory_loc=/u01/app/oraInventory  
inst_group=oinstall  
EOF  
chown root:oinstall /etc/oraInst.loc  
chmod 664 /etc/oraInst.loc
```

5.4 Kernel Parameters (/etc/sysctl.conf)

```
fs.aio-max-nr = 1048576
fs.file-max = 6815744
kernel.shmall = 2097152
kernel.shmmax = 4398046511104
kernel.shmmni = 4096
kernel.sem = 250 32000 100 128
net.ipv4.ip_local_port_range = 9000 65500
net.core.rmem_default = 262144
net.core.rmem_max = 4194304
net.core.wmem_default = 262144
net.core.wmem_max = 1048576
```

Apply: `sysctl -p`

5.5 Resource Limits (/etc/security/limits.conf)

```
grid    soft nproc    2047
grid    hard nproc    16384
grid    soft nofile   1024
grid    hard nofile   65536
grid    soft stack    10240
grid    hard stack    32768
grid    soft memlock  3145728
grid    hard memlock  3145728
```

```
oracle  soft nproc    2047
oracle  hard nproc    16384
oracle  soft nofile   1024
oracle  hard nofile   65536
oracle  soft stack    10240
oracle  hard stack    32768
oracle  soft memlock  3145728
oracle  hard memlock  3145728
```

Ensure PAM limits module: session required pam_limits.so in /etc/pam.d/login.

5.6 Preinstall Package

```
yum install -y oracle-database-preinstall-19c
```

Verify custom UIDs survived: `id grid / id oracle` should still show 54321/54322 with correct groups.

5.7 Disable firewalld, avahi, libvirtd; set SELinux permissive

```
systemctl stop firewalld avahi-daemon avahi-daemon.socket libvirtd
systemctl disable firewalld avahi-daemon avahi-daemon.socket libvirtd
ip link set virbr0 down
```

```
sed -i 's/^SELINUX=.*$/SELINUX=permissive/' /etc/selinux/config
setenforce Permissive
```

5.8 NTP (chrony)

```
echo 'OPTIONS="-x"' >> /etc/sysconfig/chronyd
systemctl enable chronyd
systemctl start chronyd
```

5.9 cvuqdisk RPM

From the extracted Grid Home (racnode1), then copy to racnode2:

```
CVUQDISK_GRP=asmadmin rpm -iv /u01/app/19.0.0/grid/cv/rpm/cvuqdisk-1.0.10-1.rpm
scp /u01/app/19.0.0/grid/cv/rpm/cvuqdisk-1.0.10-1.rpm root@racnode2:/tmp/
# on racnode2:
CVUQDISK_GRP=asmadmin rpm -iv /tmp/cvuqdisk-1.0.10-1.rpm
```

6. SSH Equivalence (grid and oracle, both nodes, both directions)

For **each** user (grid, then oracle), on **both** nodes:

```
su - grid
ssh-keygen -t rsa -N "" -f ~/.ssh/id_rsa
ssh-copy-id grid@racnode1
ssh-copy-id grid@racnode2
```

Verify all 4 combinations return the date with no password prompt:

```
ssh grid@racnode1 date
ssh grid@racnode2 date
```

Repeat identically for oracle.

7. Software Staging

7.1 Files Used

File	Purpose
V982068-01.zip	Grid Infrastructure 19.3 base installer
V982063-01.zip	Database 19.3 base installer
p39036936_190000_Linux-x86-64.zip	RU patch 39036936
p6880880_190000_Linux-x86-64.zip	OPatch replacement

7.2 Directory Layout (accessible to grid/oracle, NOT under /root)

```
mkdir -p /u01/stage/grid_install /u01/stage/db_install /u01/stage/patches
chown -R grid:oinstall /u01/stage/grid_install /u01/stage/patches
chown -R oracle:oinstall /u01/stage/db_install
chmod -R 755 /u01/stage
```

7.3 Extract GI (racnode1 only — as grid, directly into Grid Home)

```
su - grid
unzip -q /u01/stage/grid_install/V982068-01.zip -d /u01/app/19.0.0/grid
```

7.4 Extract DB software (racnode1 only — as oracle)

```
su - oracle
unzip -q /u01/stage/db_install/V982063-01.zip -d
/u01/app/oracle/product/19.0.0/dbhome_1
```

GI/DB propagate to racnode2 automatically via SSH equivalence during the installer's node-add step — no manual extraction needed on racnode2.

7.5 Environment Profiles

racnode1 — /home/grid/.bash_profile:

```
export ORACLE_BASE=/u01/app/grid
export ORACLE_HOME=/u01/app/19.0.0/grid
export GRID_HOME=/u01/app/19.0.0/grid
export ORACLE_SID=+ASM1
export PATH=$ORACLE_HOME/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$LD_LIBRARY_PATH
umask 022
```

/home/oracle/.bash_profile:

```
export ORACLE_BASE=/u01/app/oracle
export ORACLE_HOME=/u01/app/oracle/product/19.0.0/dbhome_1
export ORACLE_SID=racdb1
export PATH=$ORACLE_HOME/bin:$PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:$LD_LIBRARY_PATH
umask 022
```

racnode2 — identical, except ORACLE_SID=+ASM2 (grid) and ORACLE_SID=racdb2 (oracle).

8. Pre-Install Validation (cluvfy)

```
su - grid
cd $ORACLE_HOME
./runcluvfy.sh stage -pre crsinst -n racnode1,racnode2 -verbose | tee
/tmp/cluvfy_precrs.log
```

With 8 GB RAM per node, the Physical Memory check now passes without needing an override. Expected remaining benign items (hosts-file-only environment, no real DNS):

- resolv.conf Integrity — FAILED (expected, no DNS server)
- DNS/NIS name service — FAILED (expected, same cause)
- Single Client Access Name (SCAN) — may show a soft warning about DNS-based round robin; the multi-line /etc/hosts approach in Section 1.3 satisfies GI's actual requirement

All other checks (packages, users/groups, kernel params, storage, network connectivity, NTP, SSH equivalence) should show **PASSED**.

9. Grid Infrastructure Installation

9.1 Replace OPatch first

```
su - grid
cd /u01/app/19.0.0/grid
mv OPatch OPatch_old_bak
unzip -q "/u01/stage/patches/p6880880_190000_Linux-x86-64.zip" -d
/u01/app/19.0.0/grid/
$ORACLE_HOME/OPatch/patch version
```

9.2 Extract the RU patch to a staging folder

```
cd /u01/stage/patches
unzip -q p39036936_190000_Linux-x86-64.zip -d
/u01/stage/patches/GI_RU_39036936/
```

9.3 Launch gridSetup.sh (GUI, via X11)

With 8 GB RAM the memory prereq passes natively, so `-ignorePrereqFailure` is only needed if the DNS-related warnings still block progress in your installer build:

```
cd $ORACLE_HOME
./gridSetup.sh -applyRU /u01/stage/patches/GI_RU_39036936/39036936
```

Key wizard screens

Screen	Setting
Configuration Option	Configure a new cluster
Cluster Configuration	Configure an Oracle Standalone Cluster
Grid Plug and Play	Cluster Name: racnode-cluster; SCAN Name: racscan ; SCAN Port: 1521; Configure GNS: unchecked
Cluster Node Information	Add racnode2 — Public Hostname: racnode2.local, Virtual Hostname: racnode2-vip.local; run SSH connectivity test (passes instantly)
Network Interface Usage	ens192 = Public , ens224 = Private
Storage Option	Use Oracle Flex ASM for storage
Create ASM Disk Group	Disk group name: GRID ; Redundancy: External ; select only the 16 GB disk (/dev/asm-grid01); uncheck "Configure Oracle ASM Filter Driver" (udev already manages permissions)
Root script execution	Run root.sh automatically or manually on both nodes as prompted

Screen	Setting
Prerequisite Checks	Fix any "Fixable: Yes" items via Fix & Check Again; DNS/resolv.conf failures are expected and safe to bypass

9.4 Post-Install Verification

```
crsctl stat res -t
crsctl check cluster -all
crsctl query crs activeversion
```

All local/cluster resources should show ONLINE/STABLE on both nodes; CRS/CSS/EVM online on both.

If the installer's final Cluster Verification step fails purely due to SCAN/DNS resolution (racscan.local ... FAILED) while `crsctl stat res -t` confirms the cluster is genuinely healthy, it is safe to click **Skip** and close the wizard.

10. Create the DATA ASM Disk Group

```
su - grid
asmca
```

Disk Groups tab → Create → Name: **DATA** → Redundancy: **External** → select the 50 GB disk (/dev/asm-data01) → OK.

Confirm:

Disk Group Name	Size (GB)	Redundancy	State
DATA	50.00	EXTERN	MOUNTED (2 of 2)
GRID	16.00	EXTERN	MOUNTED (2 of 2)

11. Oracle Database 19c Software Install (software-only)

```
su - oracle
cd $ORACLE_HOME
./runInstaller
```

Screen	Setting
Configuration Option	Set Up Software Only
Database Installation Option	Oracle Real Application Clusters database installation
Node Selection	racnode1, racnode2 (SSH test passes)
Database Edition	Enterprise Edition
Installation Location	Auto-filled from \$ORACLE_HOME / \$ORACLE_BASE
Operating System Groups	dba / oper / asmdba as detected
Root scripts	Run root.sh when prompted, both nodes

11.1 Apply RU Patch to DB Home (both nodes, as root)

Replace OPatch in the DB Home first (racnode1):

```
su - oracle
cd $ORACLE_HOME
mv OPatch OPatch_old_bak
unzip -q "/u01/stage/patches/p6880880_190000_Linux-x86-64.zip" -d
$ORACLE_HOME/
exit
```

Apply the RU (as root, racnode1):

```
cd /u01/app/oracle/product/19.0.0/dbhome_1/OPatch
./opatchauto apply /u01/stage/patches/GI_RU_39036936/39036936 -oh
/u01/app/oracle/product/19.0.0/dbhome_1
```

Repeat on racnode2 — opatchauto for the DB Home does *not* auto-propagate; it must be run separately on each node. Copy the staged patch content across first (rsync is more reliable than scp for large trees):

```
mkdir -p /u01/stage/patches # on racnode2, if not present
rsync -avz --progress /u01/stage/patches/GI_RU_39036936
root@racnode2:/u01/stage/patches/
```

Then replace OPatch on racnode2's DB Home and run the same opatchauto apply command there.

Expected result on both nodes: patches 39034528 and 39039430 applied successfully; 39055473, 39107825, 39107855 skipped (GI/ASM-only, not applicable to rac_database target type — expected, not an error).

12. Database Creation (DBCA)

```
su - oracle
dbca
```

Screen	Setting
Database Operation	Create a database
Creation Mode	Advanced Configuration
Deployment Type	Database type: Oracle RAC database ; Configuration type: Admin Managed
Template	General Purpose or Transaction Processing
Nodes Selection	racnode1, racnode2
Database Identification	Global database name: racdb.local ; SID Prefix: racdb (yields racdb1/racdb2, matching the env profiles in Section 7.5); Create as Container database: checked, 1 PDB, name pdb

Screen	Setting
Storage Option	Automatic Storage Management (ASM); Database files location: +DATA/{DB_UNIQUE_NAME}; Use Oracle-Managed Files: checked
Fast Recovery Option	Unchecked (limited storage — 50 GB DATA disk group shared with datafiles); Enable archiving: unchecked for lab use
Database Options	Uncheck Sample Schemas to save space
Configuration Options → Memory	Use Automatic Shared Memory Management; SGA: 2048–2560 MB, PGA: 768–1024 MB with 8 GB RAM/node (headroom improved vs. the original 6 GB plan — see Section 13.6 for exact reasoning)
Management Options	Configure EM database express: checked, port 5500
User Credentials	Single password for SYS/SYSTEM/PDBADMIN, meeting Oracle complexity rules — record this password
Creation Option	Create database: checked

DBCA then runs: Prepare → Copy database files → Create/start instances on both nodes → Cluster database views → Complete DB creation → Create PDBs → Post-configuration actions. Expect **45 minutes to 1.5 hours** total depending on host load.

12.1 Result

- Global Database Name: racdb.local
- SID Prefix: racdb → instances racdb1 (racnode1), racdb2 (racnode2)
- Server Parameter File: +DATA/RACDB/PARAMETERFILE/spfile...
- EM Express URL: https://racscan.local:5500/em

13. RAC Administration Reference

13.1 Understanding SID vs. Service Name vs. Database Name

Term	Value in this build	Scope
Database Name (DB_NAME)	racdb	Whole database (shared by all instances)
DB_UNIQUE_NAME	racdb	Unique identifier in a single-cluster setup (matters more with Data Guard)
Instance SID	racdb1 (racnode1), racdb2 (racnode2)	One per instance — local to a node , used for ORACLE_SID and background process naming
Global Database Name	racdb.local	DB_NAME + DB_DOMAIN, used in tnsnames.ora/connect strings
Default Service	racdb.local	Auto-created service matching the DB name — load-balances across both instances

Term	Value in this build	Scope
PDB Service	pdb	Service name for the pluggable database — this is what applications actually connect to

Rule of thumb: You `sqlplus / as sysdba` locally using the **SID** (`ORACLE_SID=racdb1`). Applications and remote connections use the **service name** (`pdb` or `racdb.local`) via SCAN, never the SID directly — this is what allows Oracle to transparently load-balance and fail over across nodes.

13.2 Connecting

Local, node-specific (uses SID):

```
export ORACLE_SID=racdb1
sqlplus / as sysdba
```

Remote / application style, via SCAN and service name (recommended):

```
sqlplus system/<password>@racscan.local:1521/pdb
```

Sample `tnsnames.ora` entry (client-side):

```
PDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = racscan.local)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = pdb)
    )
  )
```

13.3 Cluster-Wide Status Checks (`crsctl`)

Purpose	Command
Full resource status, all nodes	<code>crsctl stat res -t</code>
Cluster health summary	<code>crsctl check cluster -all</code>
Check local CRS stack	<code>crsctl check crs</code>
CRS active version	<code>crsctl query crs activeversion</code>
CRS software version	<code>crsctl query crs softwareversion</code>
List voting disks	<code>crsctl query css votedisk</code>
Check OCR integrity	<code>ocrcheck</code>

13.4 Cluster / Database / Instance Start-Stop (`srvctl` and `crsctl`)

Run as **grid** for clusterware-level commands, as **oracle** (or `grid`, both can) for database-level `srvctl` commands.

Whole cluster stack (all nodes)

Stop CRS stack on Local node

```
crsctl stop crs
```

Stop CRS stack on Local node, force

```
crsctl stop crs -f
```

Start CRS stack on Local node

```
crsctl start crs
```

Stop/start CRS on a specific remote node (as root)

```
crsctl stop crs -n racnode2
```

```
crsctl start crs -n racnode2
```

Stop entire cluster (all nodes at once, as root)

```
crsctl stop cluster -all
```

Start entire cluster (all nodes at once, as root)

```
crsctl start cluster -all
```

Database (all instances, cluster-aware)

Status

```
srvctl status database -d racdb
```

Start database (all instances)

```
srvctl start database -d racdb
```

Stop database (all instances)

```
srvctl stop database -d racdb
```

Start/stop a single instance

```
srvctl start instance -d racdb -i racdb1
```

```
srvctl stop instance -d racdb -i racdb2
```

Database configuration details

```
srvctl config database -d racdb
```

ASM

```
srvctl status asm
```

```
srvctl start asm
```

```
srvctl stop asm
```

Listener

```
srvctl status listener
```

```
srvctl start listener
```

```
srvctl stop listener
```

SCAN and SCAN Listener

```
srvctl status scan
srvctl status scan_listener
srvctl config scan
```

Full stack shutdown/startup order (planned maintenance)

Shutdown (top-down):

```
srvctl stop database -d racdb
srvctl stop listener
srvctl stop scan_listener
srvctl stop scan
srvctl stop asm
crsctl stop cluster -all
crsctl stop crs # per node, or use -f if resources are stuck
```

Startup (bottom-up) — normally automatic on boot, but for manual recovery:

```
crsctl start crs # per node
crsctl start cluster -all
srvctl start asm
srvctl start scan
srvctl start scan_listener
srvctl start listener
srvctl start database -d racdb
```

In practice, `crsctl start crs` brings up ASM, listener, SCAN, and the database automatically in the correct dependency order — the granular commands above are mainly for targeted troubleshooting.

13.5 Managing Services (Application-Level Load Balancing / Failover)

Services let you control which instances handle which application workloads, independent of the raw database/PDB name.

Create a new service tied to the PDB, active on both instances

```
srvctl add service -d racdb -pdb pdb -s racapp \
  -preferred racdb1,racdb2 -policy AUTOMATIC
```

Start the service

```
srvctl start service -d racdb -s racapp
```

Stop the service

```
srvctl stop service -d racdb -s racapp
```

Status

```
srvctl status service -d racdb
```

Modify preferred/available instances

```
srvctl modify service -d racdb -s racapp -preferred racdb1 -available racdb2
```

Remove a service

```
srvctl remove service -d racdb -s racapp
```

Applications should connect using the **service name** (racapp or pdb), never the instance SID — this is what enables Oracle to transparently redirect sessions if a node fails.

13.6 Memory Tuning Reference (8 GB RAM/node)

With the RAM upgrade from 6 GB to 8 GB, there is now real headroom. Suggested allocation per node:

Component	Approx. RAM
OS + GI daemons (CRS/CSS/EVM/ONS/Listener)	0.8 – 1.2 GB
ASM instance	1.0 – 1.5 GB
DB instance (SGA + PGA)	2.5 – 3.5 GB
OS page cache / headroom	remainder

Recommended DBCA memory settings at 8 GB RAM: **SGA 2048–2560 MB, PGA 768–1024 MB** (vs. the tighter 1536/512 MB used under the original 6 GB plan). Adjust live via:

```
ALTER SYSTEM SET sga_target=2560M SCOPE=SPFILE SID='racdb1';  
ALTER SYSTEM SET pga_aggregate_target=1024M SCOPE=SPFILE SID='racdb1';
```

(Repeat per-instance with the matching SID, or omit SID= to apply cluster-wide if using a shared SPFILE default.)

13.7 Common Diagnostic Commands

Purpose	Command
Alert log location (per instance)	/u01/app/oracle/diag/rdbms/racdb/racdb1/trace/alert_racdb1.log
CRS/GI alert log	/u01/app/grid/diag/crs/racnode1/crs/trace/alert.log
Tail DB alert log live	tail -f /u01/app/oracle/diag/rdbms/racdb/racdb1/trace/alert_racdb1.log
Cluster-wide health check	cluvfy comp health -n racnode1,racnode2
Check interconnect	oifcfg getif
ASM disk status	asmcmd lsdk (run as grid, asmcmd shell)
ASM disk group free space	asmcmd lsdg

13.8 Adding/Removing a Node

Adding a third node later follows the same pattern used for racnode2: prepare OS/users/storage/SSH equivalence on the new node, then:

```
# From an existing node, as grid
cd $ORACLE_HOME/addnode
./addnode.sh -silent "CLUSTER_NEW_NODES={racnode3}" \
  "CLUSTER_NEW_VIRTUAL_HOSTNAMES={racnode3-vip}"
```

Followed by running `root.sh` on the new node and extending the Oracle Home via `addnode.sh` under `$ORACLE_HOME` (DB Home) similarly.

14. Quick Reference — Daily Operations Cheat Sheet

```
# Is everything up?
crsctl stat res -t
srvctl status database -d racdb

# Connect as DBA Locally
export ORACLE_SID=racdb1
sqlplus / as sysdba

# Connect via service (recommended for apps)
sqlplus system/<password>@racscan.local:1521/pdb

# Planned full shutdown
srvctl stop database -d racdb
crsctl stop cluster -all

# Planned full startup
crsctl start cluster -all
srvctl start database -d racdb

# Check Listener/SCAN
srvctl status listener
srvctl status scan

# Check ASM disk groups
su - grid -c "asmcmd lsdg"

# Tail alert logs
tail -f /u01/app/oracle/diag/rdbms/racdb/racdb1/trace/alert_racdb1.log
```

15. Summary of Assumptions Made in This Guide

- SCAN expanded from 1 IP to 3 IPs (.244, .247, .249)

- RAM increased from 6 GB to **8 GB per node**, meeting Oracle's official Grid Infrastructure 19c minimum and removing the need for the memory prerequisite override used in the original build.
- Name resolution remains **/etc/hosts-based** (no DNS server) — functional for GI/RAC but will show benign warnings during cluvfy/installer prerequisite checks; this is expected and does not block cluster operation.
- Cluster name, SID prefix, disk group names, and PDB name reflect the choices made during the actual build session (racnode-cluster, racdb, GRID/DATA, pdb)

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